

Distributed Systems: Lab 1

Client-Server Database

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1 Introduction

In the previous lab, you built a non-distributed system composed of a number of standalone machines, and each such machine was responsible for both the server and client components of your fortune database. You observed that the private copies of the data maintained by the machines could easily become inconsistent as there were no attempts of synchronization. The goal of the current lab is to mitigate this problem by utilizing the client-server model [1] of distributed systems. Specifically, we shall designate the server role to one separate machine and turn the rest into pure clients. This scenario is depicted in Figure 1; compare it with the one from Lab 0 [2].

2 Communication Protocol

In order for the server and clients to be able to interact with each other, they need to agree upon a communication protocol. In this programming project, the communication protocol has the following specification.

Each message is encoded in the JSON format [3, 4] and sent as a single line, which ends with a new-line character. Whenever a client wants to invoke a function on the server side, the client sends a message to the server using the following scheme:

```
{
  "method": method_name,
  "args": method_arguments
}
```

The `method` field contains a string with the name of the function that the client wants to call, and the `args` field contains a list of arguments that that function takes. For example, in order to add a new fortune into the database of the server, a client might send the following message:

```
{
  "method": "write",
  "args": ["Take it easy"]
}
```

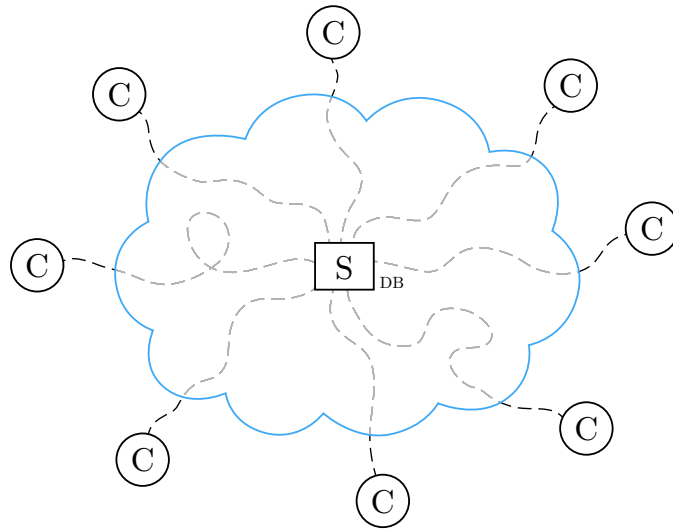


Figure 1: A distributed database with a single server (S), maintaining a database (DB), and a number of clients (C).

meaning that the method to invoke is called “write,” and “Take it easy” should be passed as the first and the only argument of that method. As noted earlier, it is important to send each message as a single line terminated by a new-line character; make sure you follow this convention when writing into the socket.

If a method invocation is successful, the server sends an acknowledgment to the corresponding client using the following scheme:

```
{
  "result": method_result
}
```

where the **result** field contains the output of the executed function encoded in the JSON format; the whole message is a single line. Note that the presence of this field is sufficient to conclude that the call was successful, and the value of the field can be **null** if the corresponding function returns nothing.

If a method invocation fails due to an error, the server replies with a message having the following structure:

```
{
  "error": {
    "name": error_class_name,
    "args": error_arguments
  }
}
```

The occurred error is encoded in **error** in such a way that it can be instantiated and raised on the client side. In fact, it is one of the requirements to your implementation.

2.1 A note on exceptions

At this point, it is worth recalling that an exception in `Python` is an instance of some exception class. In order to create such an instance, one has to know the name of the exception class and the arguments that the class' constructor requires. With this in mind, the `name` field delivers the class name of the occurred exception while `args` delivers a list of the parameters to be passed to the constructor of the class. *In our simplified case, the exception class should be a new class inheriting from `BaseException`, and with the right name.*

3 Your Task

3.1 Preparation

Continue working with the same code base that you have been working with so far, including all the changes that you have made. The files relevant to this lab are listed below. You should read and understand them.

- `lab1/client.py` — the `Client` application (should be modified);
- `lab1/server.py` — the `Server` application (should be modified);
- `lab1/test.sh` — a shell script that you can use for testing;
- `lab1/dbs/fortune.db` — the same as for Lab 0 [2];
- `modules/Common/wrap.sh` — an auxiliary script needed for `test.sh`;
- `modules/Server/lock/readWriteLock.py` — the class utilized by `Server` for the purpose of synchronization as motivated in Section 3.4. Initially this does nothing. (should be modified);
- `modules/Server/database.py` — the same as for Lab 0 [2] (should contain your previous changes).

Carefully read the source code and understand what it is doing. Pay attention to the missing parts marked with “Your code here.”

3.2 Running the code

In order to run the code, open two terminal windows and change the current folder to `lab1`. In one of the terminals, issue the following command in order to run the `Server` application (recall that the server is unique in this lab):

```
$ ./server.py
Listening to: c-204-13.eduroam.liu.se:45725
Press Ctrl-C to stop the server...
```

The code randomly chooses a port number to listen to and prints it out along with the hostname of the server machine. In the example given above, the hostname is `c-204-13.eduroam.liu.se`, and the port number is `45725`. As you have already discovered by studying the code in `server.py`, you can choose a port number yourself by specifying the corresponding argument in the command line. Now, let us create an instance of the `Client` application by running the following command in the second terminal:

```
$ ./client.py -i c-204-13.eduroam.liu.se:45725
Choose one of the following commands:
  r      :: read a random fortune from the database,
  w <FORTUNE> :: write a new fortune into the database,
  h      :: print this menu,
  q      :: exit the application.

Command> w Keep it simple.
Command> r
None
Command>
```

Note that you have to specify the address that your server is using, and it can be a different address each time you restart the server. Regardless of the correctness of the address, the system does not work at this stage since neither **Server** nor **Client** has the corresponding communication part implemented.

In order to ease the testing procedure of this and the future labs, we have implemented a set of shell scripts that open the needed terminals and run the corresponding commands for you. These scripts are called `test.sh` and can be found in the main folder of each lab. For instance, the test script for the current lab is `lab1/test.sh`. It is important to note, however, that you should be able to set up everything yourself without the help of these scripts. So, have a look at the content of the scripts and make sure you understand what they do. You can also modify these scripts as you wish.

3.3 A note on manual testing

We use a fairly simple plaintext protocol, where you can easily play the part of server or client to test out functionality.

Try communicating manually with the server or client part using “nc”¹. This will allow you to see how the client tries to communicate with the server, try out sending malformed data to the client or server to see how they respond, or respond in a way that should raise an exception at the client side.

As an exercise, start two terminals, one that listens continuously/repeatedly (with “-k”) to TCP port 1234. Then try connecting to it and communicating between the terminals (break with Ctrl-C).

```
$ nc localhost 1234
Hello
^C
```

Additionally, try out

```
$ echo '{"method" : "read"}' | nc localhost 1234
```

The command `ss`² can be useful for figuring out which ports are open (and which process uses them, if you have root on your system).

¹See <https://wizardzines.com/comics/netcat/> or <https://linux.die.net/man/1/nc>

²See <https://wizardzines.com/comics/ss/>

3.4 Communication

Using Python's sockets [5], complete the implementation of `server.py` and `client.py`. Your code should satisfy the following requirements:

- Both `Server` and `Client` should implement and follow the communication protocol described in Section 2.
- The server should process each client in a separate thread such that it does not stop serving other incoming connections. Your implementation should be thread safe as motivated below.
- The server should be robust: it should catch and adequately respond to exceptions including those due to potential network problems, violations of the protocol, invocations of non-existing functions, invocations of proper functions but with wrong arguments, etc.
- Whenever a client receives an error from the server, it should instantiate and raise it as if the error has occurred locally. The user of the `Client` application should not experience any difference between the non-distributed and distributed versions of the system. Thus, if there is a `ServerLocalPermissionError` on the server side, the traceback on the client side should contain something like `ServerLocalPermissionError: This is the error text` and *not* `Exception: ServerLocalPermissionError: This is the error text`

That is, you can't just raise a standard `Exception` with the right text and write `'ServerLocalPermissionError'` in it. It must be an exception type called `ServerLocalPermissionError`. As is most often the case, you should *not* use `eval`.

Note that the Python language provides function, `type`, for (among other things) dynamic class creation (or, to be more precise, type object creation).[6]. See above for a discussion of what we mean by raising the exception locally.

3.5 Synchronisation

Since the server can have several threads serving client requests in parallel, the operations on the database issued by the clients can interfere with each other and leave the database in a corrupted state. In order to prevent this behavior, you should protect the critical parts of your code with a read/write lock provided to you in `readWriteLock.py`. Consider where the locking should take place and how to make it efficient in terms of the waiting time.

Lastly, make sure that, in spite of all possibly occurred exceptions, the server keeps on working until it is explicitly stopped by the user. The clients may crash, but the server must keep working.

3.6 Additional questions

When demonstrating this lab (as with any lab in the course), you should be able to explain your code, but also to evaluate it, and relate it to course concepts. In many cases this will involve discussing design choices and simplifications that

have been made for you, and may be somewhat speculative. In some cases you might be asked to answer questions with (at least in retrospect) quite obvious answers. In this lab, this might include questions such as

- What happens if a message is lost, in this design?
- In this lab (series) we raise exceptions with the right name, that inherit from “BaseException“, in order to make it seem like we are using a local object. Are there any pros and cons of this approach? Are there limits to this way of making remote exceptions seem local (more on this in the next lab)?
- One cannot focus on everything, and this lab (and course) abstracts away many security concerns. It should however be mentioned. What security concerns can be raised about our current system? Who can access it, and from where?
- We use the ReadWriteLock on the server side.. Why don't we provide this on the client side instead? Would you notice any difference when doing manual testing with the textual user interface?

We do not require written answers to this, and you might not need to provide a solution to all issues raised by the questions.

4 Conclusion

In this lab, you have implemented a rather naïve distributed system: there is only one server and each client is required to know the current address of the server. The former issue makes our distributed system both unreliable (e.g., your carefully written code will not protect the server against a blackout) and inefficient (e.g., if there are many clients, the workload of a single server machine can be substantially high making clients wait for a response for a long time). The second issue makes it inconvenient for the clients to connect to the server as the hostname and/or port of the server might easily change, and the clients need to keep track of such changes. This inconvenience will become especially bothering when you allow your distributed system to have several servers; therefore, in the next lab, you will address this problem first.

References

- [1] <https://www.ida.liu.se/~TDDD25/lectures/lect2.pdf>.
- [2] <https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab0.pdf>.
- [3] <https://en.wikipedia.org/wiki/JSON>.
- [4] <https://docs.python.org/3/library/json.html>.
- [5] <https://docs.python.org/3/library/ipc.html>.
- [6] <https://docs.python.org/3/library>.